Code No. 8408

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) IV-Semester (MAIN) Examinations July-2025 1005237290O8

COMPUTER ORGANIZATION AND MICROPROCESSORS

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	What are the main components of any General Purpose computer.	2	1	1
b)	What is the main role of a micro-program sequencer.	2	2	2
c)	Explain the importance of PUSH & POP instruction.	2	2	3
d)	Examine the importance of 8259 programmable interrupt controller.	2	4	4
e)	List the difference between microprocessor and microcontrollers.	2	1	5
f)	Write format of single and double precision format.	2	4	1
2. a)	Explain the common bus system architecture in detail, and provide a neatly labeled block diagram.	6	2	1
b)	Explain the RTL (Register Transfer Language) representation for arithmetic operations such as addition or subtraction.	6	2	1
3. a)	Describe memory-reference instructions and their format in a typical instruction set.	6	3	2
b)	Draw the flowchart for Instruction Cycle and explain it.	6	4	2
4. a)	Describe different instruction formats with examples.	6	3	3
b)	Explain the structure and functioning of cache memory with relevant terms like hit ratio and miss.	6	2	3
5. a)	Draw and explain the internal architecture of the 8085 microprocessor mentioning the function of each block.	6	4	4

	b) Explain the architecture and function of the Intel 8257 DMA controller.	6	2	4
6.	a) Discuss the advantages and limitations of different 8-bit microcontroller families with respect to system design and performance.	6	3	5
	b) Explain the process of interfacing an ADC with a microcontroller, highlighting the necessary circuit connections and programming steps.	6	4	5
7.	a) What are the examples of Logic and shift micro-operations.	6	2	1
	b) Explain the programming techniques for 8051 microcontrollers, including the use of assembler directives and subroutine handling.	6	2	5

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